

## **Unnamed\_Unit**

### **Unit 5 - Knowledge Check**

#### **PDF Documents**

##### **PDF Document 1: 1738213053450-Chapter 4 - Unit 5 - Knowledge Check.pdf**

This PDF document is included in the unit materials.

# KNOWLEDGE CHECK

## 1 What is sustainability?

- ☐ A The ability to keep using resources without depleting them
- ☐ B A way to increase consumption of natural resources
- ☐ C The process of building faster technology
- ☐ D A method to prevent technological failure

CORRECT ANSWER ☒ A

## 2 Why is sustainability important for the environment?

- ☐ A It helps reduce the use of renewable energy
- ☐ B It ensures that resources are available for future generations
- ☐ C It increases the rate of industrial production
- ☐ D It prevents the development of new technology

CORRECT ANSWER ☒ B

## 3 What does a flood alarm system do?

- ☐ A Monitors the wind speed during storms
- ☐ B Alerts people when there is a risk of flooding
- ☐ C Reduces water levels in rivers
- ☐ D Tracks rainfall throughout the year

CORRECT ANSWER ☒ B

## 4 Which of the following is a key component of a flood alarm system?

- ☐ A Temperature sensor
- ☐ B Light sensor
- ☐ C Motion detector
- ☐ D Water level sensor

CORRECT ANSWER ☒ D

## 5 Which of the following is a technological solution for flood prevention?

- ☐ A Solar power
- ☐ B Flood alarm systems
- ☐ C Wind turbines
- ☐ D Electric vehicles

CORRECT ANSWER ☒ A

## 6 What is the purpose of a loop in programming?

- ☐ A To run a section of code multiple times
- ☐ B To stop the program
- ☐ C To check if a condition is true
- ☐ D To store large amounts of data

CORRECT ANSWER ☒ A

## 7 What programming concept is essential for creating a flood alarm system?

- ☐ A Random loops
- ☐ B Conditional statements
- ☐ C Variables
- ☐ D Classes and objects

CORRECT ANSWER ☒ B



## Critical Thinking Challenge

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- 1 Why is it important to use renewable energy sources, such as solar power, for systems like flood alarms, and how could this benefit disaster-prone areas?

- 2 How could conditional statements and loops in programming be used together to make a flood alarm system more reliable and efficient?

## Expanding Knowledge on the Topic



### WHAT IS THE ENVIRONMENT?

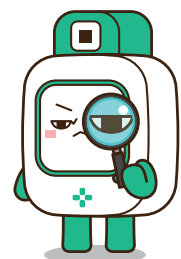
The environment is everything around us—air, water, land, plants, and animals. It is important because it provides us with what we need to live, like clean air to breathe and water to drink. But our environment is facing challenges like pollution and climate change, which are damaging the Earth. This is where sustainability comes in!

### WHAT IS SUSTAINABILITY?

Sustainability means using resources in a way that ensures future generations can also enjoy them. It's about taking care of the planet so that we don't run out of the things we need, like clean water, energy, and healthy ecosystems. By making small changes, we can live more sustainably and reduce the negative impact on the environment.

### WHY DOES SUSTAINABILITY MATTER?

Sustainability is important because our actions today affect the world of tomorrow. If we continue to waste resources or harm the environment, future generations might not have the same opportunities to thrive. For example, using too much plastic can lead to pollution in the oceans, harming marine life. If we don't take action, these problems will only get worse.





## HOW CAN WE BE MORE SUSTAINABLE?

There are many ways to live sustainably. Here are a few simple things you can do to help the environment:



- **Reduce, Reuse, Recycle:** These three actions can reduce the amount of waste we produce.
- **Save Energy:** Turn off lights and electronics when you're not using them.
- **Use Less Water:** Be mindful of how much water you use when brushing your teeth or taking a shower.
- **Plant Trees:** Trees help clean the air and provide homes for animals.
- **Use Reusable Bags:** Instead of plastic bags, use cloth bags when shopping.

## HOW TECHNOLOGY HELPS WITH SUSTAINABILITY

Technology plays a big role in helping us protect the environment. Solar panels, for example, harness the sun's energy to create electricity without polluting the air. Electric cars produce less pollution than regular cars, and new inventions are constantly being developed to help clean our air and water.

## WHAT CAN YOU DO?

Even as students, you can make a difference. Learning about sustainability and making eco-friendly choices in your daily life can help protect the planet. Every small action counts, and together, we can create a healthier environment for ourselves and future generations.

